

JARVIS on your phone

A local LLM that listens and talks back

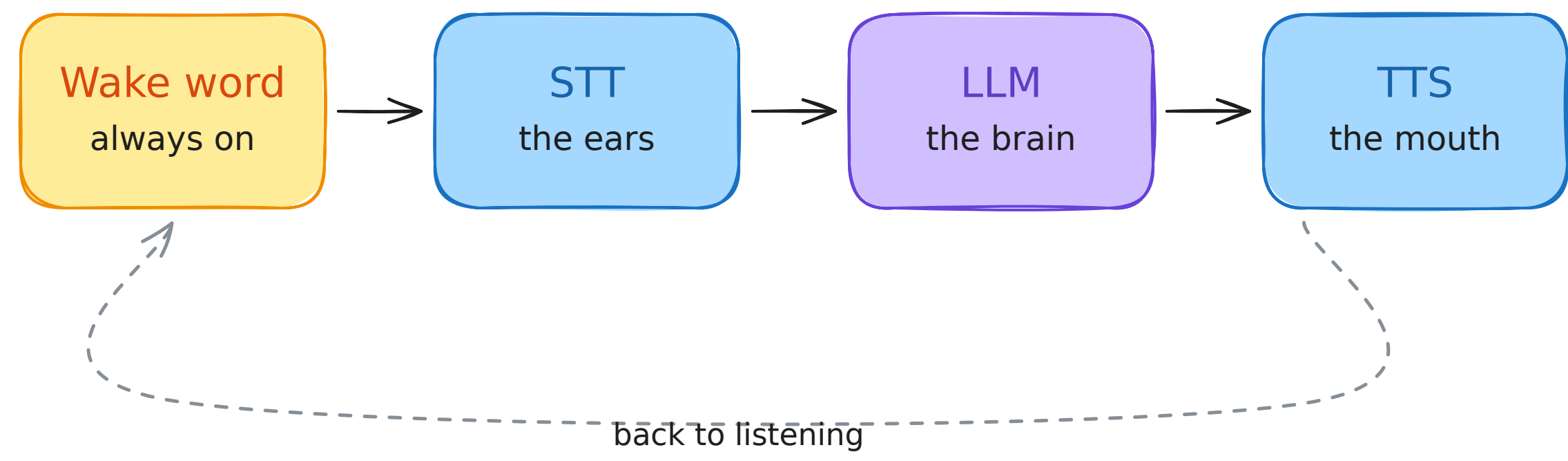
An LLM, a voice and a wake word on Android
Airplane mode on. Nothing leaves the phone.



Dishant Gandhi · @droidcity_

THE BIG PICTURE

One loop, all on the phone

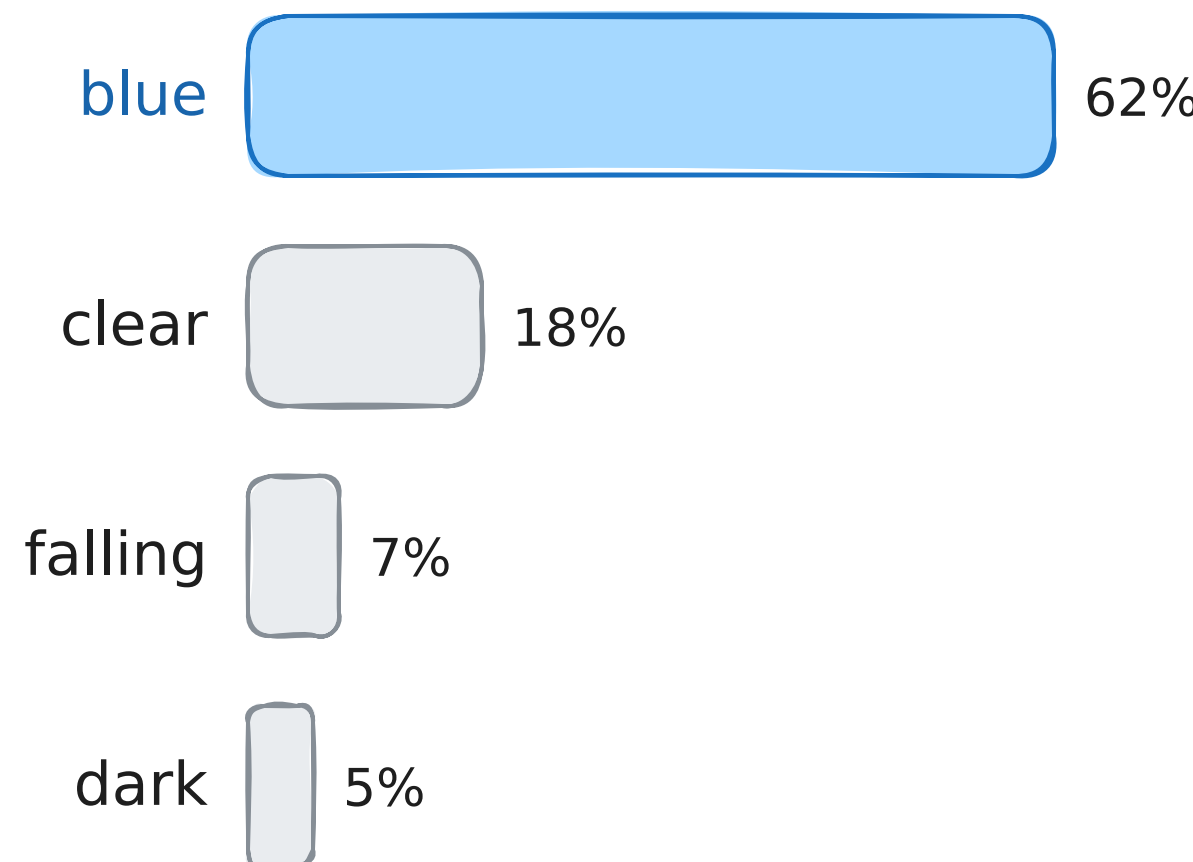


If any of these four needs the internet, it isn't really your assistant.
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HOW LLMs WORK

It guesses the next word

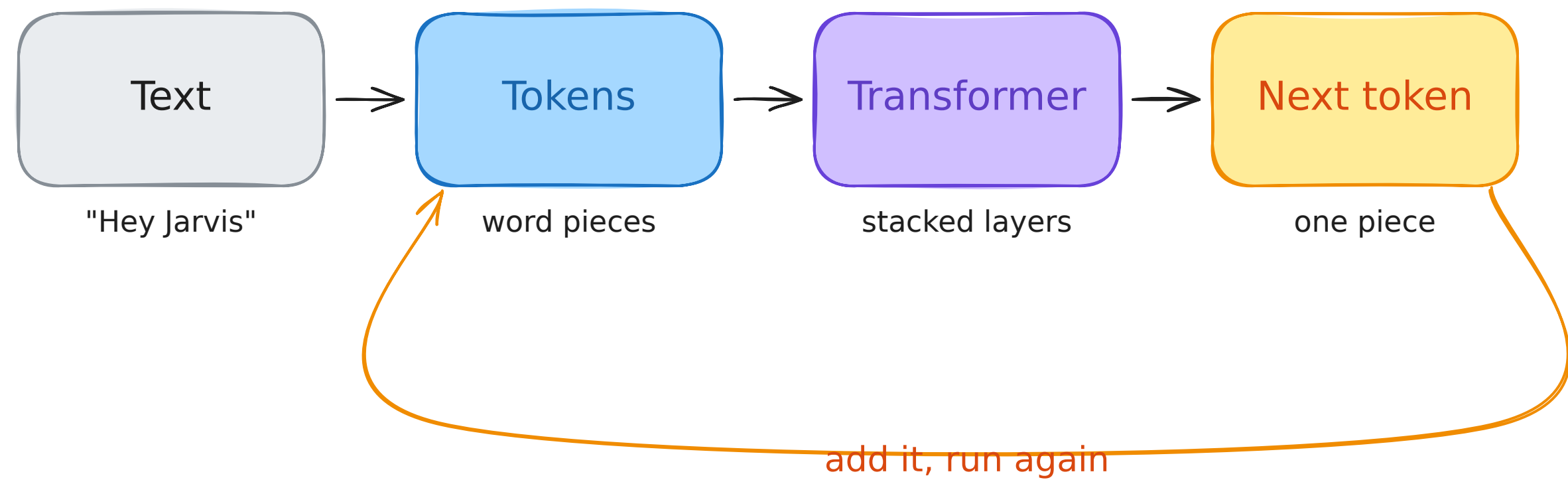
The sky is _____
example numbers



No magic, no thinking. It's a very, very good autocomplete.
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HOW LLMs WORK

Then it does it again



A 200-word reply is a few hundred laps of this loop. Speed is everything.
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HOW LLMs WORK

A model is just numbers

$$4B \times 4 \text{ bits} \approx 2 \text{ GB}$$

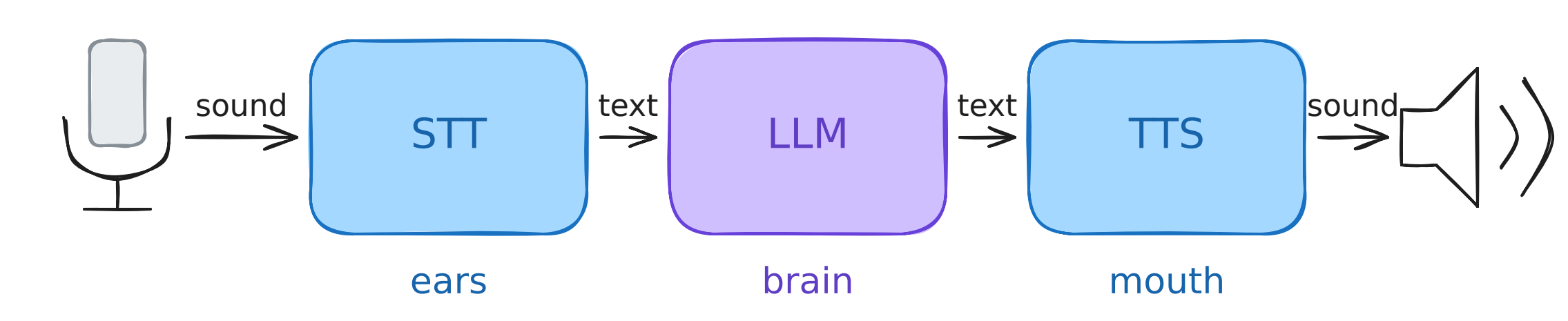
weights each on disk

This one multiplication decides whether a model fits in your pocket.
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Slide 06 — Ears brain mouth

STT AND TTS

Ears, brain, mouth

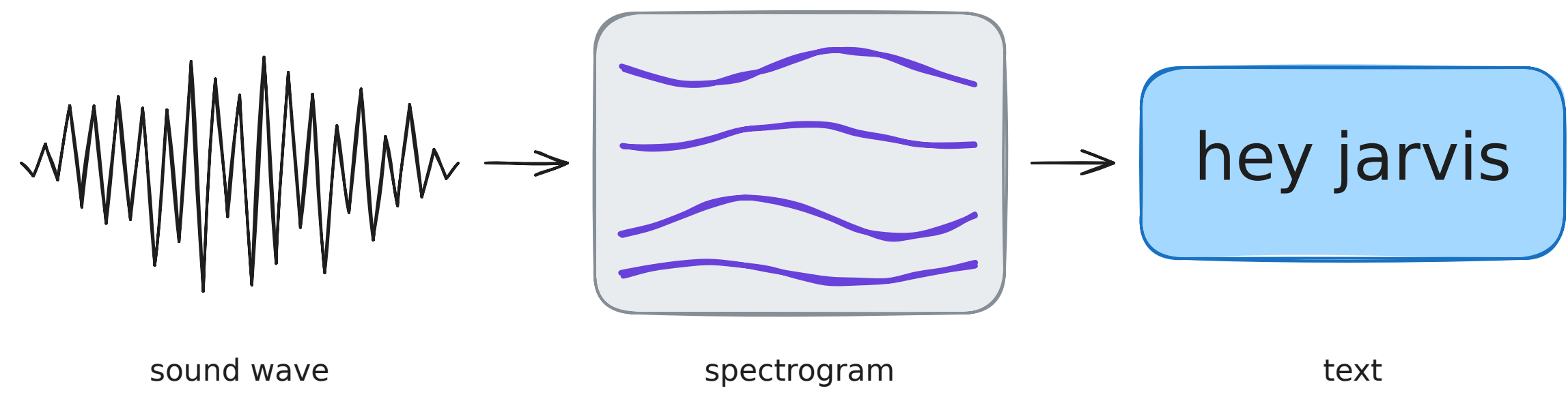


The LLM never hears your voice. It only ever reads text.
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Slide 07 — Speech to text

SPEECH TO TEXT

Sound becomes a picture, then words

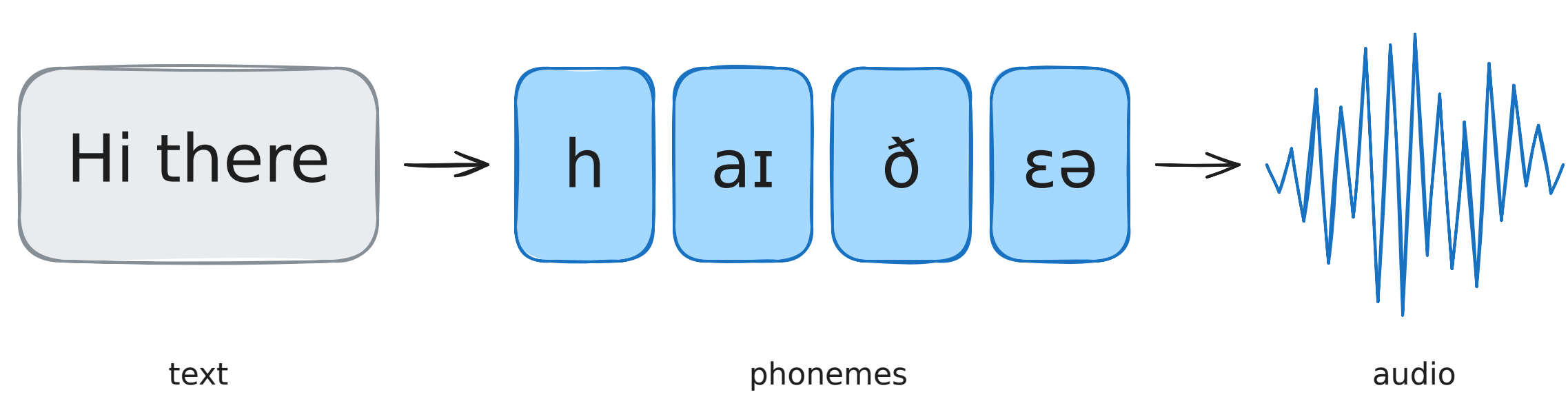


The model reads a picture of your voice, not the raw sound.
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Slide 08 — Text to speech

TEXT TO SPEECH

Words become sounds, then a voice

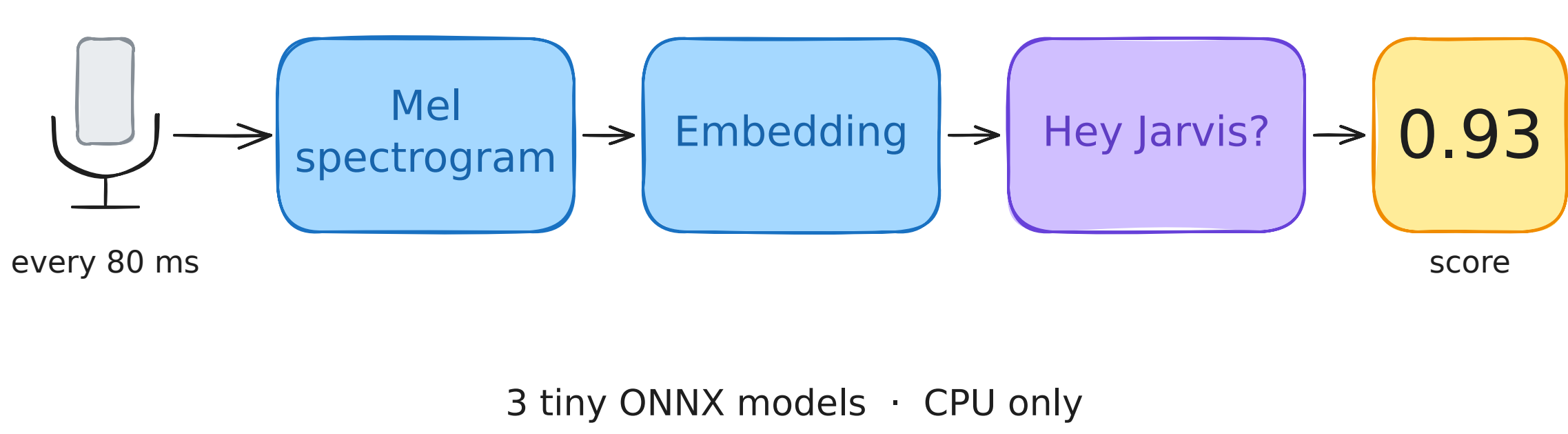


Same trick in reverse: the model paints the sound wave directly.
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Slide 09 — Wake word pipeline

WAKE WORD

Always listening, barely working

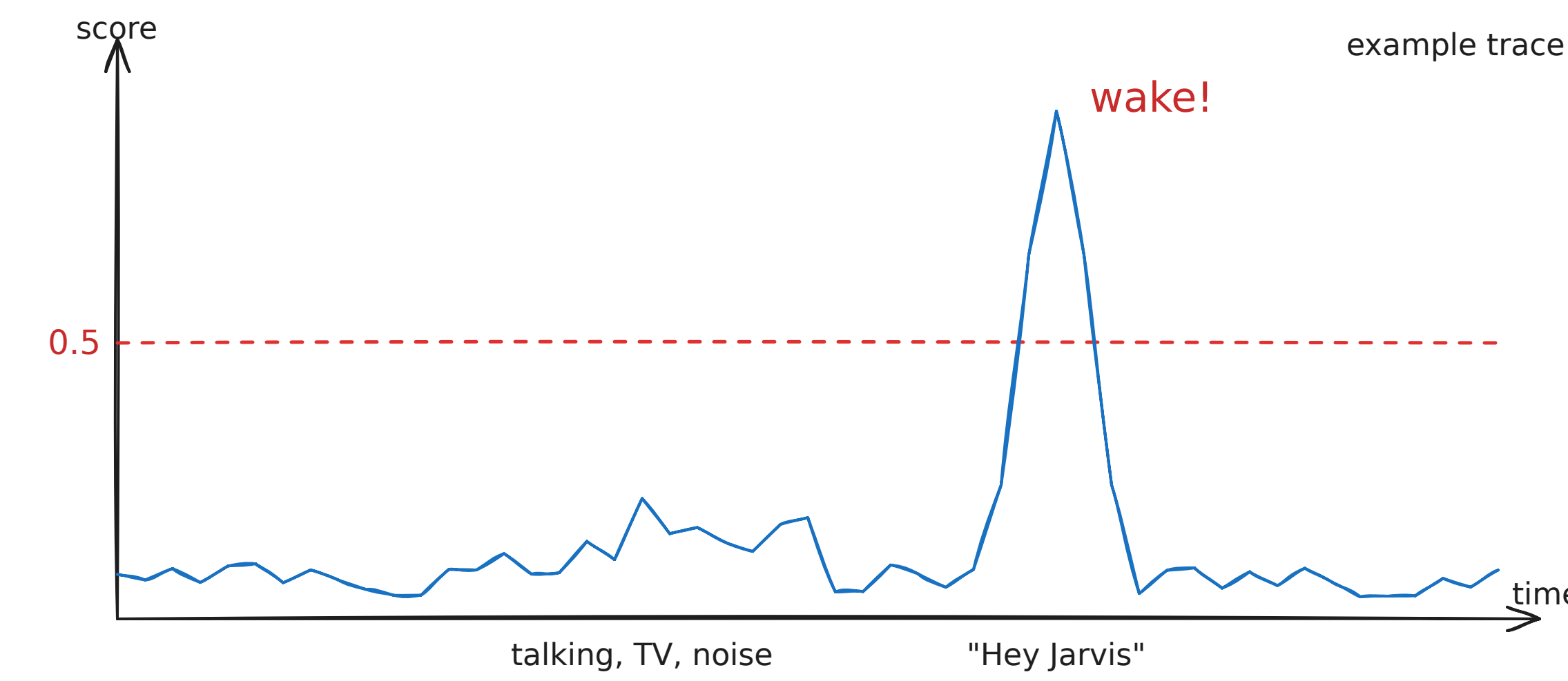


Three tiny models do the boring job all day so the big one can sleep.
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Slide 10 — Wake threshold

WAKE WORD

Cross the line, wake up

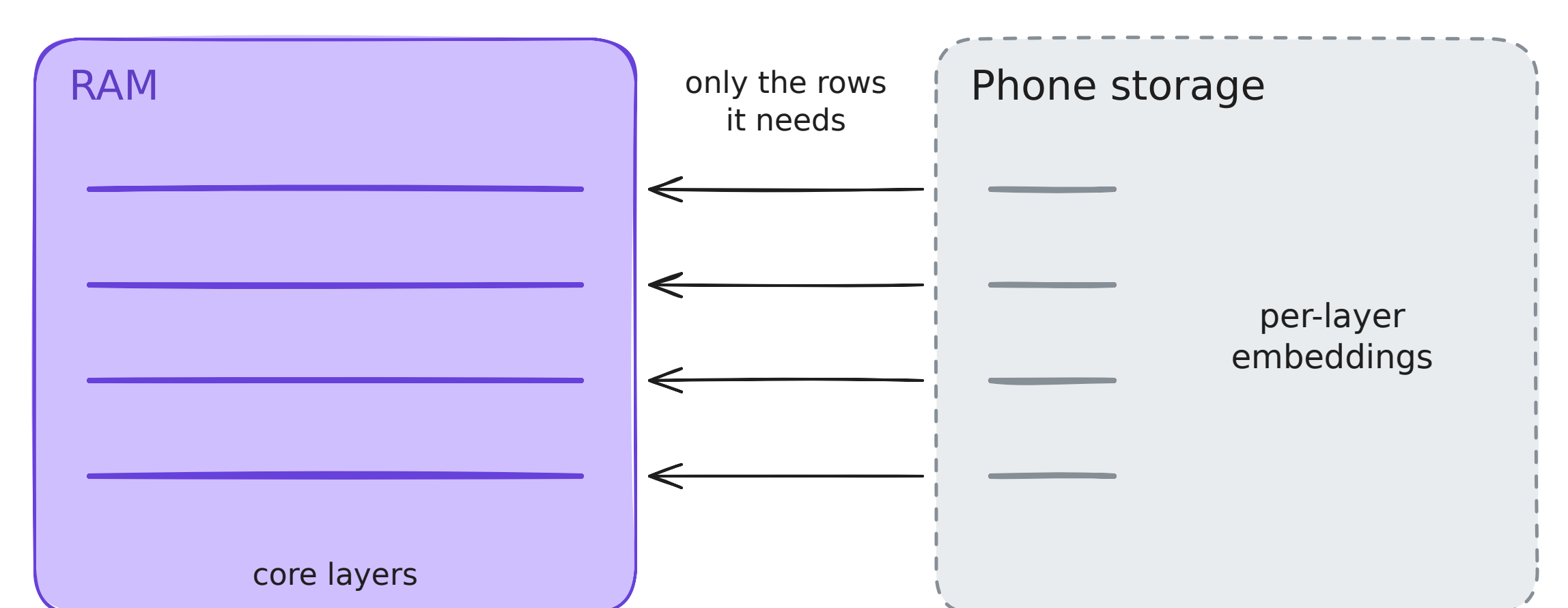


Set the bar too low and your TV starts talking to Jarvis.
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Slide 11 — Gemma effective

WHY GEMMA FLIES

The E means effective

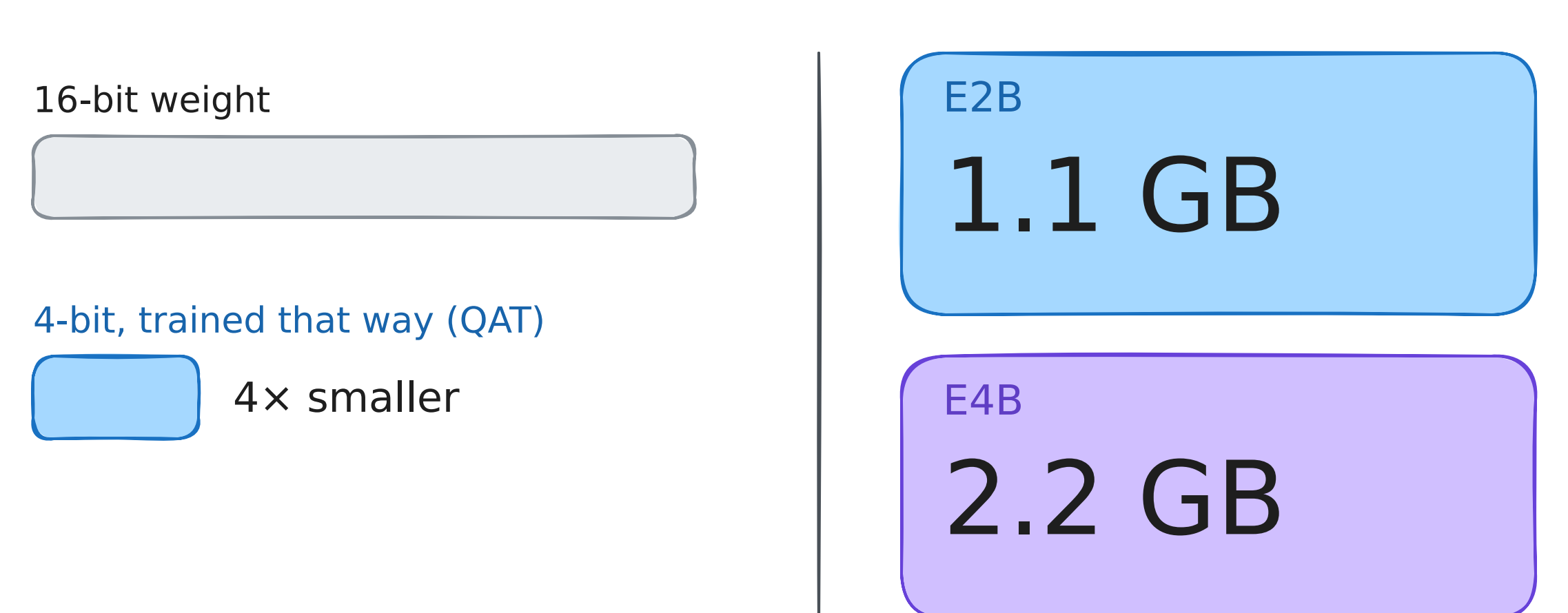


It stores more than it loads. That's the whole trick.
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Slide 12 — Gemma QAT

WHY GEMMA FLIES

Trained to be small

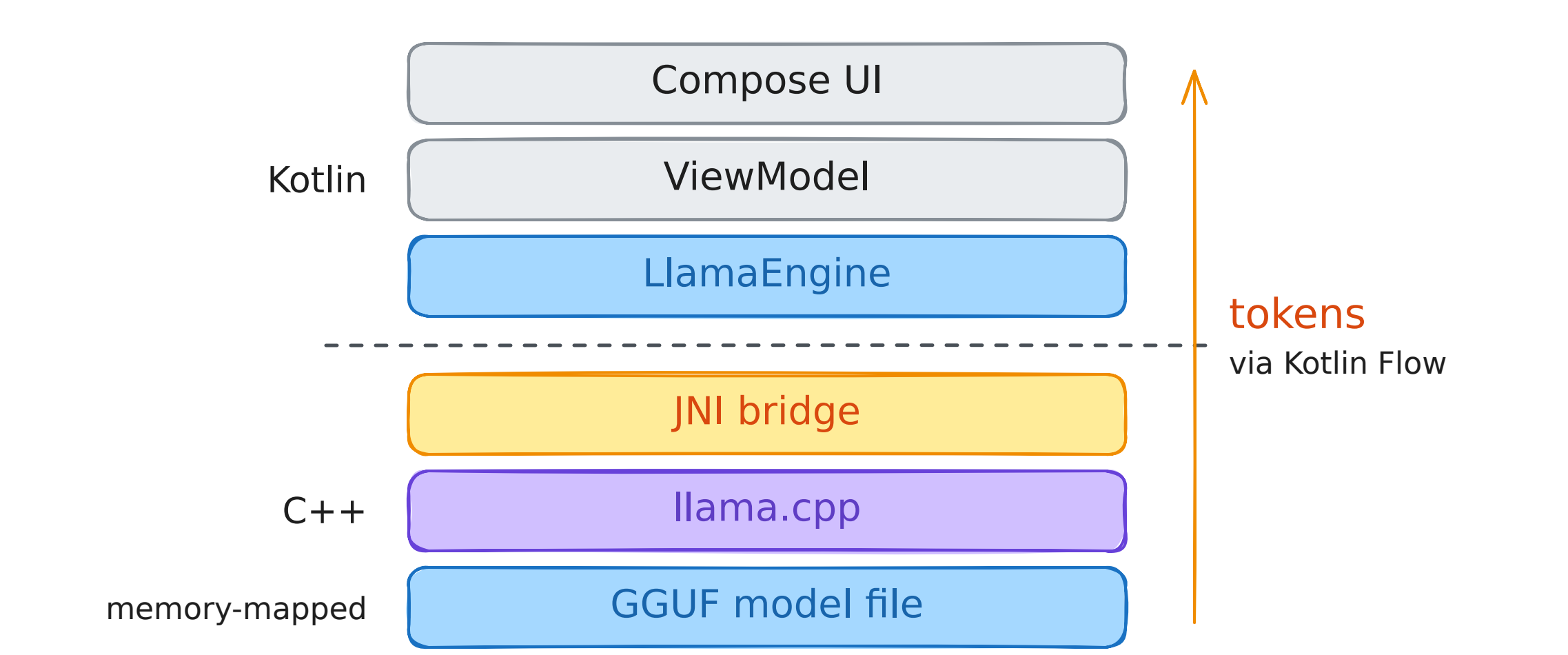


Squash it after training and it gets dumber. Train it small and it mostly doesn't.
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Slide 13 — Integration stack

LLM IN THE APP

From tap to tensor

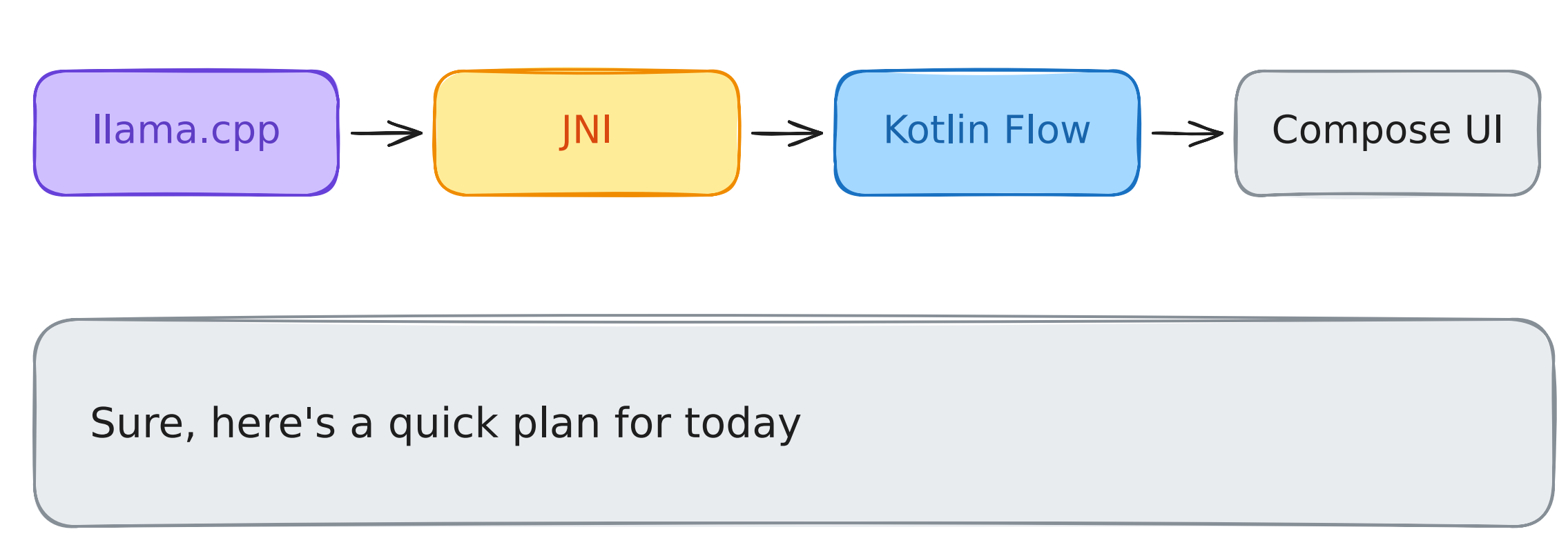


Kotlin never touches the weights. It asks nicely through JNI.
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Slide 14 — Streaming

LLM IN THE APP

Show words as they come

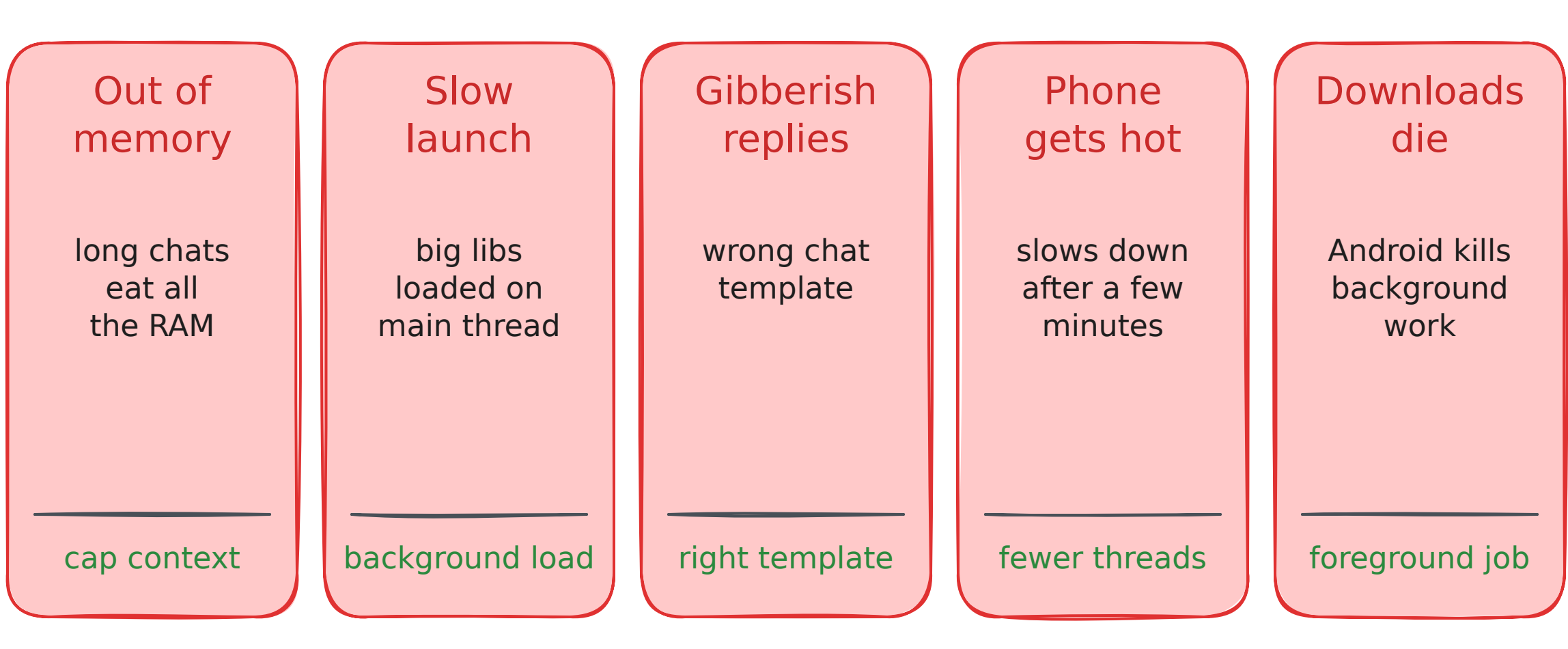


Nobody minds slow if they can watch it think.
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Slide 15 — What broke

WHAT BROKE

Five ways it fell over

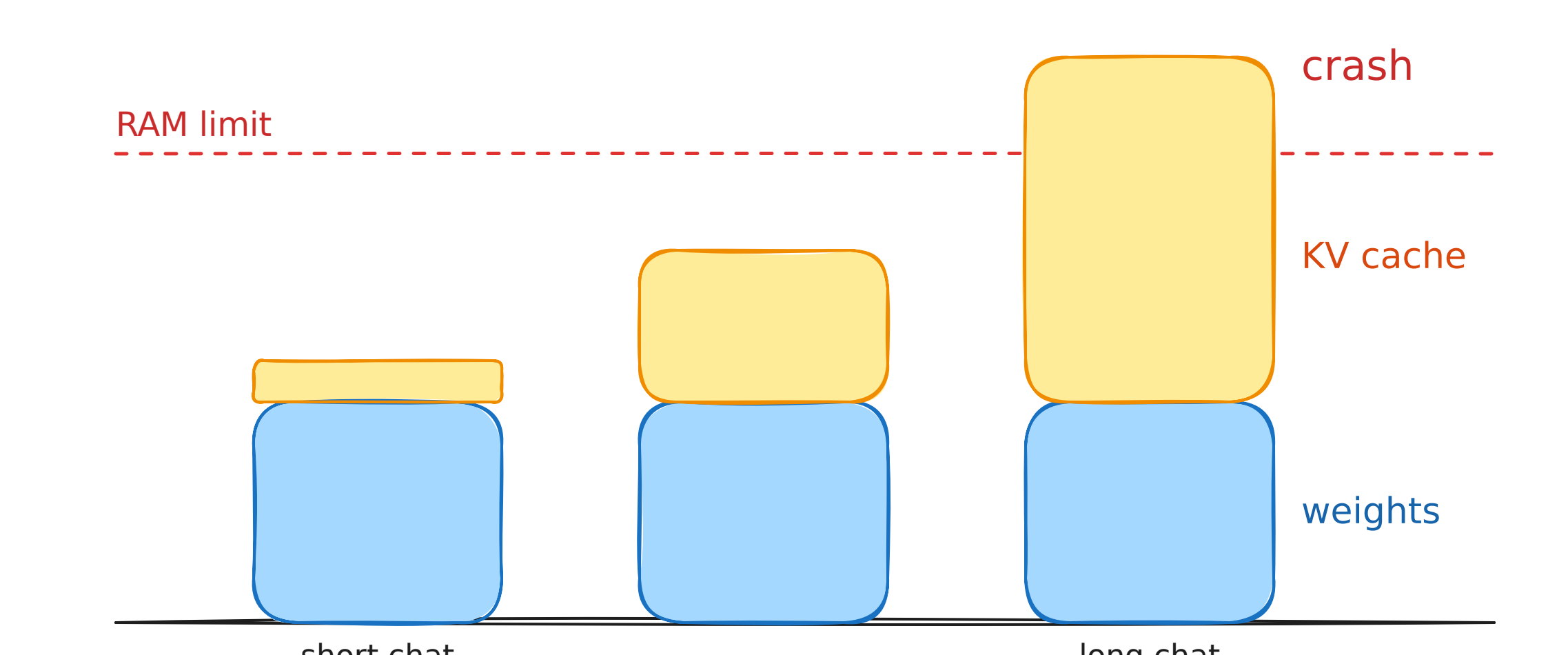


Every one of these crashed on a real phone before it made this list.
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Slide 16 — Memory growth

WHAT BROKE

Every word costs memory

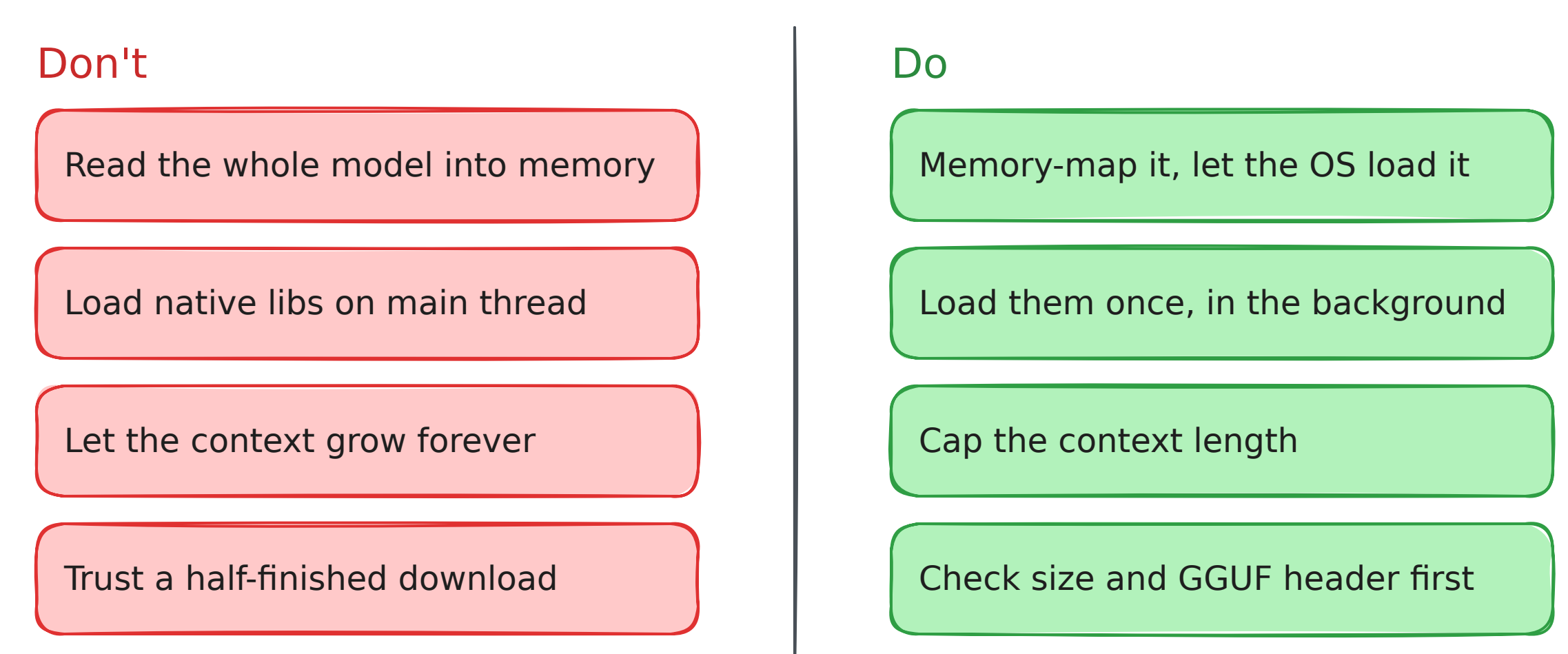


The model doesn't kill your app. The conversation does.
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Slide 17 — Tips and tricks

WHAT FIXED IT

Don't, and do

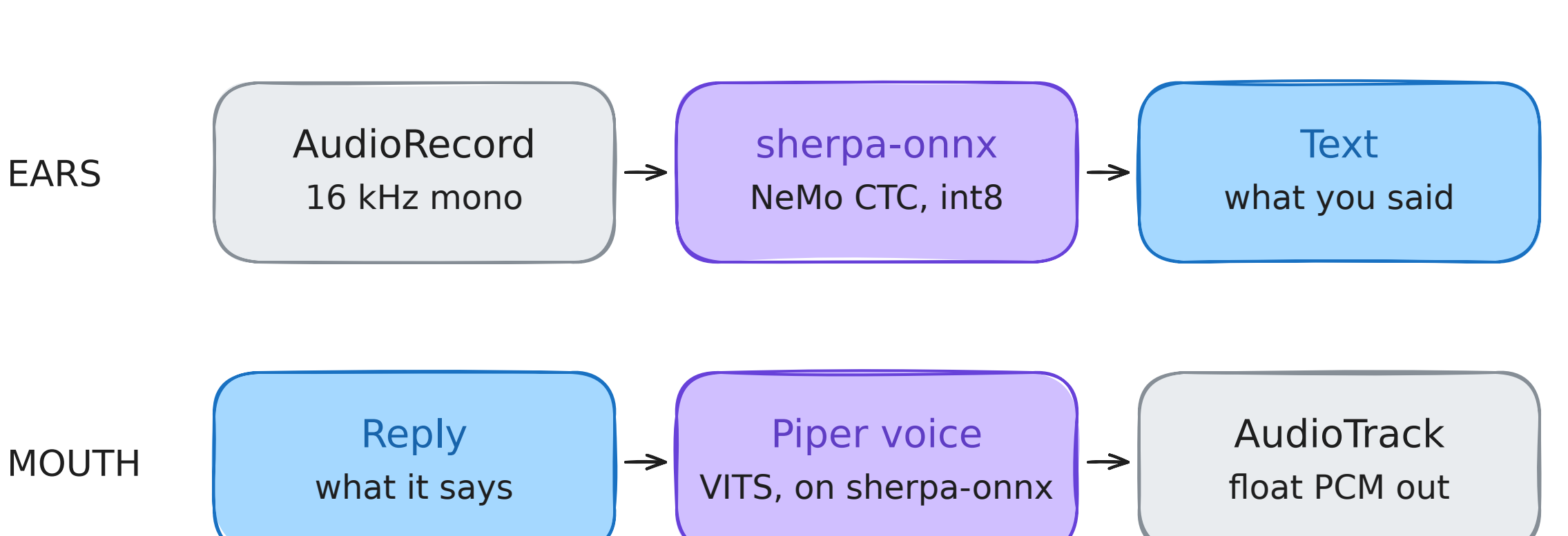


Treat the phone like it's already low on memory, because it is.
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Slide 18 — Voice on Android

VOICE ON ANDROID

Voice, fully on-device

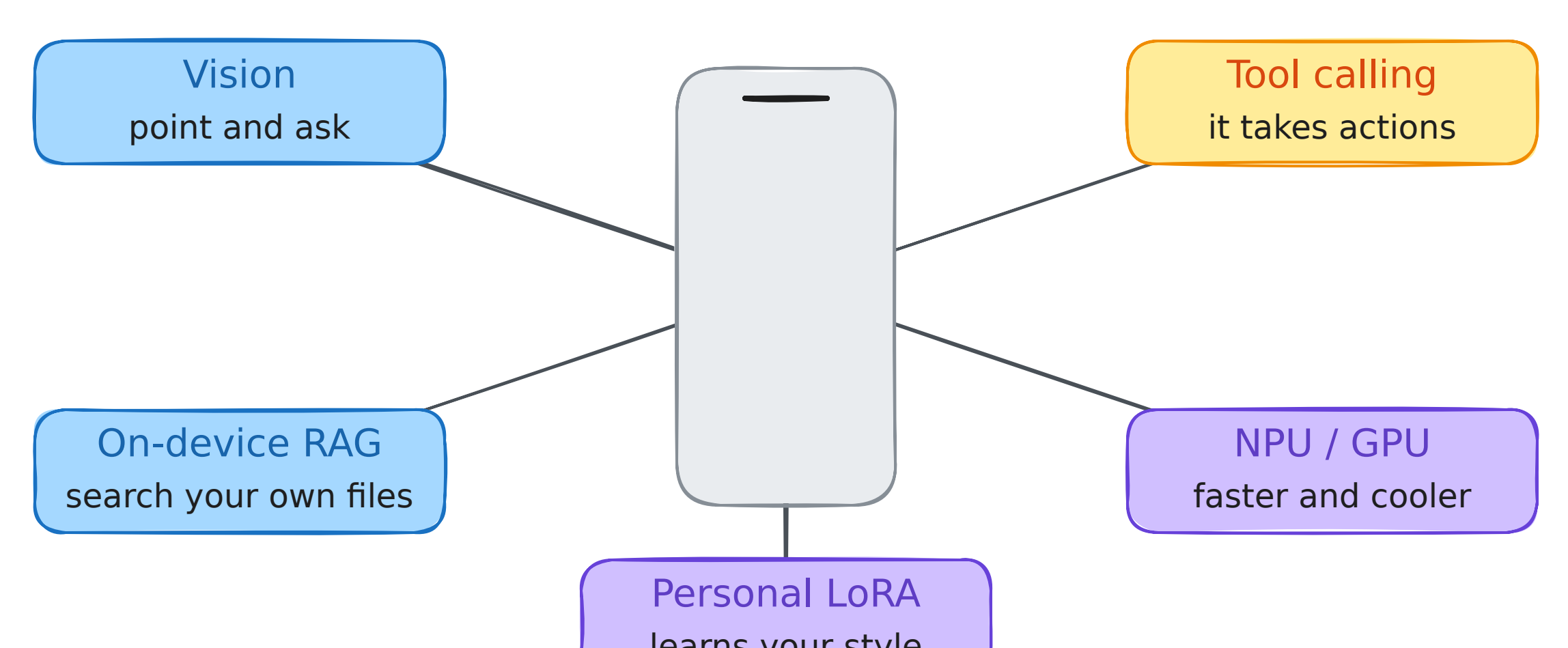


English today · Hindi built in, switched off for now
Two ONNX models, zero network calls, no API bill.
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Slide 19 — Future

WHAT'S NEXT

Where this goes next



Tool calling is the one that turns a chatbot into an assistant.
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Slide 20 — Demo and QA

Demo

Say "Hey Jarvis"

Airplane mode is still on. Then questions.

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